

DATA ENGINEERING · BUSINESS INTELLIGENCE

MSBI Masterclass: SSIS, SSAS & SSRS

A topic-complete practical masterclass through SQL Server Integration Services (SSIS), Analysis Services (SSAS), and Reporting Services (SSRS). Build enterprise ETL pipelines, multidimensional analytical cubes, tabular models, and paginated dashboards.

THE COMPLETE MICROSOFT DATA PLATFORM

Master the end-to-end flow: Ingest and clean messy transactional data with SSIS ETL, structure analytical cubes with SSAS, and distribute parameterized executive reports with SSRS.

SSIS ETL

75+Transformations &
tasks

SSAS CUBES

15+Multidimensional &
Tabular

SSRS REPORTS

08+

Enterprise report types

FORMAT

100%

Practical live coding

[Explore Course Webpage →](#)[Talk to MSBI Mentors →](#)

END-TO-END PIPELINE ARCHITECTURE

The Enterprise *Microsoft BI Ecosystem*

Understand how raw transactional databases are extracted, transformed, modeled, and visualized through the unified Microsoft BI stack.

TIER 1 • ETL

01

SSIS (Integration Services)

High-throughput data extraction, cleansing, transformation, and warehouse loading:

- **Control Flow:** Loop containers, Execute SQL.
- **Data Flow:** 20+ built-in transformations.
- **SCD Engine:** Type 1 & Type 2 historical data.
- **Deployment:** SSISDB & SQL Agent automation.

TIER 2 • MODELING

02

SSAS (Analysis Services)

Multidimensional cubes and modern in-memory Tabular analytical models:

- **Cubes & DSVs:** Star and Snowflake schemas.
- **Hierarchies & KPIs:** Drill-downs & status trends.
- **Tabular Models:** VertiPaq memory engine.
- **DAX & MDX:** Calculated measures & time intelligence.

TIER 3 • REPORTING

03

SSRS (Reporting Services)

Pixel-perfect paginated reports, dynamic charts, and automated distribution:

- **Layout Types:** Table, Matrix pivot & Charts.
- **Parameters:** Cascading & multi-select filters.
- **Drill-Through:** Master-detail subreports.
- **Subscriptions:** Scheduled email & file share.

AUDIENCE

Who This Is Designed For

SQL Developers, Database Administrators, Data Analysts, and QA Engineers wanting to transition to Microsoft BI & Data Engineering.

SQL Developers

Data Analysts

ETL Engineers

PREREQUISITES

What You Need Before Starting

Basic SQL querying (SELECT, JOIN, GROUP BY). All SSIS, SSAS, SSRS, Data Warehousing, and SSDT concepts are taught from scratch.

Basic SQL

SSDT Tools

No Prior BI Required

THE ENTERPRISE ADVANTAGE

Master the data platform powering Fortune 500 decision systems.

Learn Data Warehousing · Build SSIS Pipelines · Design Multidimensional Cubes · Publish Paginated SSRS Reports with automated distribution.

Ready to master the Microsoft BI stack? View batch schedules and enroll

[View Course Webpage →](#)

PHASE 01 & 02 • HIGH-THROUGHPUT ETL

SSIS Architecture, *Control Flow & Transformations*

Master Data Warehousing fundamentals, Control Flow tasks, loop containers, 20+ transformations, SCD, and SQL Agent deployment.

CONTROL FLOW

01

Tasks, Containers & Precedence

Orchestrating complex ETL task workflows and looping logic:

- **Core Tasks:** Execute SQL, File System, Send Mail, Script Task (C#), FTP Task.
- **Containers:** Sequence Containers, For Loop & Foreach Loop (Files & ADO objects).
- **Precedence Constraints:** Success, Failure, Completion, and Expression rules.

DATA FLOW

02

Data Flow Engine & Buffers

High-speed in-memory data pipeline execution:

- **Sources & Destinations:** OLE DB, ADO.NET, Flat File, Excel, XML, Raw Files.
- **Buffer Tuning:** DefaultBufferMaxRows, DefaultBufferSize, and thread management.
- **Data Viewers:** Live pipeline debugging and grid inspection.

TRANSFORMATIONS

03

20+ Built-in Transformations

Cleansing, shaping, routing, and aggregating streaming data:

- **Row Level:** Data Conversion, Derived Column, Conditional Split, Copy Column.
- **Lookups & Joins:** Lookup (Full / Partial / No Cache), Merge, Merge Join (Inner/Left/Full).
- **Aggregations & Routing:** Aggregate, Sort, Multicast, Union All.

SCD & DEPLOYMENT

04

Slowly Changing Dimensions & Catalog

Managing dimension history and production scheduling:

- **SCD Type 1 & Type 2:** SCD Wizard, custom lookup & historical row tracking.
- **Error Handling:** Event Handlers (OnError, OnWarning), Row Redirects, Logging.
- **Deployment:** Project Deployment Model, SSISDB Catalog, SQL Agent automation.

SSIS PRACTICE LAB & CHECKPOINT: Build an end-to-end SSIS ETL pipeline extracting raw CSVs, cleansing data via Derived Column & Lookup transformations, implementing SCD Type 2 dimension loads, and scheduling automated runs via SQL Server Agent.

Have questions about SSIS packages? Connect with our mentors

[Schedule Mentor Call →](#)

PHASE 03 • ANALYTICAL MODELING

SSAS Multidimensional Cubes & *Tabular DAX*

Design high-performance analytical data models, dimension hierarchies, Key Performance Indicators (KPIs), and in-memory Tabular DAX models.

DATA MODELING

01

Data Source Views & Dimensions

Structuring Star and Snowflake data schemas in SSDT:

- **DSVs:** Creating Data Sources, Data Source Views, logical primary keys & calculations.
- **Dimensions:** Standard, Time, and Degenerate dimensions. Key & Name columns.
- **Attribute Relationships:** Optimization for query performance and aggregation paths.

CUBE DESIGN

02

Measure Groups & Hierarchies

Building and processing multidimensional OLAP cubes:

- **Measure Groups:** Additive, Semi-additive, and Non-additive measures.
- **User Hierarchies:** Multi-level drill-down paths (Year → Quarter → Month → Day).
- **Cube Processing:** ProcessFull, ProcessData, ProcessIndex, and automated refreshes.

CALCULATIONS & KPIS

03

Calculated Members & KPI Metrics

Writing custom business metrics and visual performance indicators:

- **MDX Expressions:** Calculated members, growth percentages, and ratios.
- **KPIs:** Value, Goal, Status, and Trend graphics (traffic lights, arrows).
- **Perspectives & Actions:** Defining focused cube slices and URL/Drill-through actions.

TABULAR & DAX

04

Tabular Models & DAX Measures

Modern in-memory column-store analytical modeling:

- **VertiPaq Engine:** In-memory column storage and relationship definitions.
- **Calculated Columns vs Measures:** Row context vs Filter context.
- **DAX Basics:** CALCULATE, FILTER, RELATED, SUM, and Time Intelligence (YTD/MTD).

SSAS PRACTICE LAB & CHECKPOINT: Build a complete Retail Sales Analysis Cube in SSDT with 3 Dimensions, a Date Hierarchy, 4 Measure Groups, Custom KPI indicators with status graphics, and deploy to an Analysis Services instance.

PHASE 04 • ENTERPRISE VISUALIZATION

SSRS Reporting & *End-to-End Capstone*

Build pixel-perfect paginated reports, dynamic matrix pivots, cascading parameters, automated subscriptions, and connect the end-to-end BI pipeline.

REPORT AUTHORIZING

01

Datasets, Tables & Matrix Pivots

Designing professional enterprise report layouts:

- **Data Sources & Datasets:** Shared vs Embedded data connections.
- **Data Regions:** Table, Matrix (Pivot), List, and Chart visualizations.
- **Expressions:** Dynamic formatting, IIF logic, conditional background styling.

INTERACTIVITY & DELIVERY

02

Parameters, Drill-Downs & Subscriptions

Adding user interactive controls and automated delivery:

- **Parameters:** Multi-value, default values, and Cascading dropdown filters.
- **Interactive Navigation:** Drill-down toggle visibility, Document Maps, Subreports.
- **Report Manager Portal:** Security roles, Scheduled email & file-share Subscriptions.

END-TO-END CAPSTONE PROJECT

03

Enterprise Financial & Retail Data Warehouse

Connect the full Microsoft BI stack into a production-grade automated analytics pipeline:

1. SSIS ETL Engine

Ingest dirty transactional data, clean fields, apply SCD Type 2, and load Star Schema Data Warehouse.

2. SSAS Analytical Cube

Model Fact & Dim tables, user hierarchies, custom KPIs, and in-memory Tabular DAX measures.

3. SSRS Dashboards

Publish parameterized Matrix reports with drill-down, charts, and automated executive email subscriptions.

CAPSTONE VERIFICATION & CODE REVIEW: Deploy the complete automated pipeline on SQL Server, execute automated ETL refreshes, query the SSAS cube with Excel/Power BI, and schedule daily SSRS executive PDF dispatches.

The complete end-to-end Microsoft BI portfolio project

[View Capstone Details →](#)

YOUR PRACTICAL DATA BI PATHWAY

Master the tools. Build the evidence.

Explore the complete MSBI masterclass, review live batch timings, and connect with RPA Vault mentors to accelerate your data engineering career.

Explore the complete course

See the full curriculum, learning model, project path, and course details online.

rpavault.com/course/msbi-masterclass/
→

Talk to RPA Vault Mentors

Discuss your background, course fit, batch options, and enrollment steps.

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